

Causal inference from the ground up

a foundations primer (companion to the cookbook)

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One-line orientation: this is the document that comes *before* the cookbook. It builds every term the cookbook assumes — treatment, counterfactual, confounding, identification, the do-operator — from nothing, using marketing examples and analogies aimed at a physicist. No library code here; just the concepts and the words, in the order they build on each other.

1. The three nouns: treatment, unit, outcome

A **treatment** is *any action whose effect you want to measure*. The word is medical (give a patient a drug), but in business it is anything you can deliberately **do**: send a discount email, show an ad, launch a loyalty program, change a price, redesign a page. Nothing more mysterious than “the intervention.”

The thing that receives the treatment — a customer, a store, a region — is a **unit**. The thing you measure afterwards — spend, clicks, retention — is the **outcome**. We write the treatment as T (often $T = 1$ for treated, $T = 0$ for not), and the outcome as Y .

The entire subject is one question: **did T change Y , and by how much?** Everything below is about answering it honestly.

2. The counterfactual: the number you can never measure

Here is the problem that makes this a real science and not just bookkeeping. Take one customer, Anna. You email her a discount ($T = 1$) and she spends €80 ($Y = 80$). Did the email *work*?

To answer honestly you would need to know what Anna would have spent **had you not emailed her**. Call that number $Y(0)$ — her outcome *in the world where $T = 0$* . Maybe €80 (the email did nothing). Maybe €30 (it added €50). Maybe €95 (it annoyed her). You cannot know, because in this world you *did* email her. The un-emailed Anna does not exist.

So for each unit there are two **potential outcomes**:

$$Y(1) = \text{outcome if treated}, \quad Y(0) = \text{outcome if not treated},$$

and the effect *on that individual* is $Y(1) - Y(0)$. The trouble: you only ever observe **one** of the two. The one matching the treatment actually given:

$$Y = T \cdot Y(1) + (1 - T) \cdot Y(0).$$

The other is permanently missing. This is the **fundamental problem of causal inference**, and the honest one-liner is: *causal inference is the science of reconstructing the outcome you did not get to see*.

Physicist’s translation. Think of it as a branching: given its state, the system evolves down exactly one branch — the treatment you applied — and you observe that branch. The other branch ($Y(0)$ for a treated unit) is a genuine counterfactual: a well-defined quantity that was simply never realized. It is also, clearly, a **missing-data** problem — half of the table $\{Y(1), Y(0)\}$ is blank by construction. That framing will serve you well: everything after this is principled imputation of the blanks.

3. From individuals to averages

We cannot recover Anna’s missing branch. Fortunately, business decisions don’t need it — they run on **averages**. “Does emailing *this kind of customer* pay?” not “what exactly would Anna have done?”

The average is written with an **expectation** $\mathbb{E}[\cdot]$ — for a physicist this is exactly the ensemble average $\langle \cdot \rangle$, the mean over the population. The headline quantity is the **Average Treatment Effect**:

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)] = \langle Y(1) \rangle - \langle Y(0) \rangle.$$

Now the impossibility softens. We still can’t see any *individual’s* $Y(0)$, but if we can get a trustworthy estimate of the population average $\langle Y(0) \rangle$ — the average spend of comparable customers who weren’t emailed — we’re in business. The whole difficulty collapses onto one word: **comparable**.

4. The enemy: confounding (why association $\neq$ causation)

Suppose you email your *loyal* customers the discount and take your *casual* customers as the comparison (“**control**”) group. The loyal group spends more afterward. Did the email cause it?

Almost certainly not — or not entirely. Loyal customers spend more *anyway*. The difference you measured is a mix of two things: the real effect of the email, and the fact that **the two groups were different to begin with**. The variable “loyalty” drove both *who got treated* and *how much they spend*. A variable like that — a **common cause** of treatment and outcome — is a **confounder**, and it is the central villain of the whole field.

Physicist’s translation. Two detectors both responding to the same background field will read as correlated even though neither drives the other. Loyalty is the background field here; “got the email” and “spent a lot” are the two detectors. Their correlation is real but not causal. **Association is what you see in the joint distribution; causation is what survives when you remove the common drivers.**

This is why the naïve “treated group minus control group” is usually *wrong*: it silently blames (or credits) the treatment for differences that were really about *who ended up in each group* — **selection**. Defeating confounding is the single problem every technique in the cookbook solves, each in its own way.

5. The gold standard: randomization

There is one move that annihilates confounding at a stroke: decide who gets treated by a **coin flip**. Randomly assign each customer to email or no-email. This is an **A/B test** (in statistics, a **randomized controlled trial**, RCT).

Why it works: if assignment is a coin flip, then loyal and casual customers land in the treated and control groups in the *same proportions*. Every confounder — loyalty, age, past spend, ones you never even measured — is balanced across the two groups on average. Formally, randomization makes the treatment **statistically independent** of the potential outcomes:

$$T \perp \{Y(0), Y(1)\}.$$

(“ $\perp$ ” means independent: knowing T tells you nothing about what the person’s $Y(0)$ or $Y(1)$ *would* be — because it was decided by a coin, not by anything about the person.) Under that independence, the treated group’s average outcome is a fair stand-in for what the control group *would* have done, and vice versa, so

$$\text{ATE} = \langle Y \mid T = 1 \rangle - \langle Y \mid T = 0 \rangle$$

— a plain difference of observed group means. When you can randomize, causal inference is *easy*.

6. Life without randomization: identification

You often can’t randomize. You can’t randomly bankrupt regions to study a downturn; you can’t ethically deny a working treatment to half your patients; and most often the “treatment” already happened in the past with no coin flip — you just have the data as it fell (this is **observational** data).

Then you must *argue your way* to a causal answer using assumptions. Making that argument explicit is called **identification** — showing that, *if* certain assumptions hold, the causal effect is recoverable from the data you have. The standard argument rests on three assumptions; learn these, because they are the fine print behind every method:

1. **Unconfoundedness** (a.k.a. *ignorability*): once you condition on enough measured background variables X , treatment is as-good-as-random. Written $\{Y(0), Y(1)\} \perp T \mid X$. In words: *within customers who look identical on X , who got the email was effectively a coin flip*. This is the big assumption — it says you measured all the important confounders.
2. **Positivity** (a.k.a. *overlap*): at every value of X there are both treated and untreated units — $0 < P(T = 1 \mid X = x) < 1$. You can’t compare like-with-like if some kind of customer was *always* emailed and never appears in the control group.
3. **Consistency / SUTVA**: the treatment is well-defined and one unit’s treatment doesn’t spill over onto another (emailing Anna doesn’t change Bob’s spend).

The quantity $e(x) = P(T = 1 \mid X = x)$ — the probability of being treated given your background — is important enough to have a name: the **propensity score**.

The recipe these assumptions unlock is intuitive: instead of comparing the treated and control groups *as wholes* (unfair — different mixes), compare them **within strata of X** , like against like, then average the within-stratum differences back up weighted by how common each stratum is. Loyal-treated vs loyal-control; casual-treated vs casual-control; then combine. Formally,

$$\text{ATE} = \mathbb{E}_X \left[\underbrace{\mathbb{E}[Y \mid T = 1, X]}_{\text{avg spend, treated, this } X} - \underbrace{\mathbb{E}[Y \mid T = 0, X]}_{\text{avg spend, control, this } X} \right].$$

This is the **adjustment formula** (or **g-formula**). Reading it left to right: *estimate the effect within each kind of customer, then average over the kinds*. Almost every “condition on the right variables and model the outcome” method is a flavor of this.

The catch is assumption 1: it’s **untestable**. You are asserting you controlled for every important confounder. You can never prove it from data — which is why serious causal work always comes with a **sensitivity analysis**: *how strong would an unmeasured confounder have to be to overturn my conclusion?* Keep this split sharp in your head: **identification is an assumption about the world; estimation is the statistics you do once you’ve granted it**. No amount of clever modeling rescues a broken identification argument.

7. Seeing versus doing: the do-operator

Now the single most important conceptual leap, and the one where your physics training is a gift.

There is a difference between **observing** that a variable takes a value and **forcing** it to. Compare:

- *Observing* $X = x$: you look at units where X happens to equal x . But X became x for reasons, and those reasons may also affect the outcome. So conditioning on $X = x$ mixes the effect of X with the effects of whatever *drove* X to x .
- *Doing* $X = x$: you reach in and **set** $X := x$ from outside, overriding its usual causes. Now X 's value carries no information about those causes, because you, not they, set it.

Pearl writes the second as $\text{do}(X = x)$, and the causal effect is properly a statement about **do**, not about conditioning:

$$\text{effect of treatment} = \mathbb{E}[Y \mid \text{do}(T = 1)] - \mathbb{E}[Y \mid \text{do}(T = 0)].$$

Physicist's translation (this one is exact). Model the system as a set of **structural equations** — the “equations of motion” of your variables, each written as a function of its direct causes plus noise:

$$X := f_X(\text{parents of } X, u_X), \quad T := f_T(\text{parents of } T, u_T), \quad Y := f_Y(\text{parents of } Y, u_Y).$$

Conditioning on $T = 1$ means *restricting to solutions of the full system that happen to have* $T = 1$ — the coupling to T 's causes is intact. **Intervening**, $\text{do}(T = 1)$, means *deleting* T 's equation and *clamping* $T = 1$, then letting the rest of the system run. It is literally surgery on the equations. The everyday version: **reading a node voltage** (it correlates with everything the source also drives) versus **attaching an ideal voltage source to that node** (you override those couplings). Same value, completely different implied distribution over the rest of the circuit.

Under the three assumptions of §6, the surgical **do** quantity reduces to the observational adjustment formula — that reduction is what §6 buys you. When it doesn't reduce so cleanly, you need the graphical machinery next.

8. Causal graphs: reading confounders and colliders off a picture

Draw each variable as a node and each direct causal mechanism as an arrow. The result — a **causal DAG** (directed acyclic graph; “acyclic” = no variable causes itself through a loop) — is just the dependency diagram of those structural equations, and it lets you *see* what to control for.

A path from T to Y that goes **backwards** out of T (against an arrow) is a **backdoor path** — it's how confounding sneaks in. $\text{loyalty} \rightarrow \text{email}$ and $\text{loyalty} \rightarrow \text{spend}$ create the backdoor $\text{email} \leftarrow \text{loyalty} \rightarrow \text{spend}$. To get the honest effect you must **block** every backdoor path, and you block one by conditioning on (“controlling for”) a variable sitting on it — here, *loyalty*. The minimal set of variables that blocks all backdoors is exactly the X you need in the adjustment formula. That's the **backdoor criterion**, and it's the whole content of “what should I control for?”

But — and this is the counterintuitive part that trips up even seasoned analysts — **controlling for the wrong variable actively creates bias**. Two traps:

- **Mediators.** If the treatment works *through* a variable ($\text{email} \rightarrow \text{opened} \rightarrow \text{spend}$), controlling for “opened” removes part of the very effect you're trying to measure. Don't adjust for things the treatment causes.
- **Colliders.** A variable with *two* arrows pointing into it ($T \rightarrow C \leftarrow Y$) is a **collider**. Conditioning on a collider *opens* a path that was closed, manufacturing a spurious association between T and Y out of thin air.

Physicist's translation of the collider. Two independent causes feed one effect. A bulb lights if *either* of two independent switches is on. The switches are independent — until you observe the bulb. Given “bulb is on, switch A is off,” you *know* switch B is on: conditioning on the common effect makes the causes dependent. This is “**explaining away**,” and it is why conditioning on a post-treatment variable can invent a correlation that isn't causal. Blindly throwing every available column into a regression is dangerous precisely because some of those columns are mediators or colliders.

So the graph does real work: it tells you the *specific* variables to include (to block backdoors) and the ones to *exclude* (mediators, colliders). This is what pathmc and pymc-marketing’s causal layer automate in the cookbook.

9. One effect for everyone, or a different effect for each?

Two genuinely different questions hide under “the treatment effect”:

- The **average effect** (ATE): one number for the whole population. “Did the loyalty program lift revenue overall?” This is **program evaluation / incrementality**.
- The **varied effect**: the effect *depends on who you look at*. The discount barely moves loyal customers but strongly moves lapsed ones. Written as a function of the customer’s features x , this is the **Conditional Average Treatment Effect**:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x].$$

“Conditional” = for a given kind of customer; the effect is now a landscape over x , not a scalar. Because it’s called *heterogeneous* (“not the same for everyone”), you’ll see **heterogeneous treatment effects (HTE)** — the term at the center of Michele’s session.

Why HTE is the whole game for targeting: **if the effect were identical for everyone, there’d be no one to select** — you’d rationally treat all or none. It’s precisely *because* $\tau(x)$ varies that you can find the customers it works on (high $\tau(x)$) and spend only on them. Uplift modeling is nothing but estimating $\tau(x)$ and acting on it.

One more term you’ll meet: sometimes a method can only recover the effect for a *particular sub-population* — e.g. the customers an instrument actually nudges. That restricted quantity is a **Local Average Treatment Effect (LATE)**: “local” to the subgroup the method can speak about. Worth knowing so you report the right scope.

10. Where Bayesian thinking enters (your home turf)

Everything so far was about **identification** — turning a causal question into an estimable quantity, an assumption-laden argument about the world. **Estimation** — actually computing it from finite, noisy data — is where your Bayesian machinery walks in, and it fits with almost no friction.

Put a model with parameters θ on the response surface $\mu_t(x; \theta) = \mathbb{E}[Y \mid T = t, X = x; \theta]$ (or on the whole structural system). Every estimand above — ATE, $\tau(x)$ — is a **deterministic functional of θ** . So the posterior $p(\theta \mid \text{data})$ you already know how to compute *pushes forward* into a full posterior distribution over the causal effect itself:

$$p(\tau(x) \mid \text{data}) = \int \delta(\tau(x) - [\mu_1(x; \theta) - \mu_0(x; \theta)]) p(\theta \mid \text{data}) d\theta.$$

That’s the payoff: a **credible interval on the treatment effect** falls out for free — no bootstrap, no delta method — and the do-operator becomes literally “simulate the structural model forward under the intervention, propagating parameter uncertainty.” A physicist reads that integral as marginalizing a derived observable over the posterior; it’s the same move you already make.

The discipline to carry over from §6: Bayes sharpens *estimation*, never *identification*. A tight, beautiful posterior computed under a false unconfoundedness assumption is confidently wrong. Priors don’t fix broken causal arguments — the graph and the assumptions do.

11. The map: which foundation leads to which method

With the vocabulary in place, the cookbook splits cleanly into two families, and every technique is just a specific answer to “*what would have happened without the treatment?*”

Family A — you believe unconfoundedness holds given measured X (§6). Then model the outcome surface and apply the adjustment formula. The methods differ only in *how flexibly* they model it:

- **BART / Bayesian Causal Forest** (cookbook §1) — flexible $\tau(x)$ for uplift/targeting.
- **pathmc structural models** (§§2–4) — do-operator, segment effects, mediation.
- **pathmc / pymc-marketing DAG tools** (§5) — choose the adjustment set; catch colliders.
- **pymc-marketing causal MMM** (§6) — incrementality and budget under a marketing DAG.

Family B — unconfoundedness fails, but the world handed you a structure that mimics randomization. Each exploits a different natural source of “as-good-as-random” variation:

- **Synthetic control** (§7) — build the missing $Y(0)$ trajectory from untreated units (geo lift).
- **Difference-in-differences** (§8) — treated-vs-control *change* over time cancels fixed differences.
- **Regression discontinuity** (§9) — a threshold rule makes units just above/below comparable.
- **Interrupted time series** (§10) — extrapolate the pre-treatment trend as the counterfactual.
- **Instrumental variables** (§11) — a “nudge” that affects Y only through T (recovers a LATE).

Read the cookbook next with these five ideas in hand — **treatment/unit/outcome; the unseeable counterfactual; confounding; identification via adjustment or via a natural experiment; average vs heterogeneous effects** — and every section will read as a named tool rather than new vocabulary.

The eight words to walk in with

Treatment (the intervention) · **counterfactual / potential outcome** (the branch you didn’t observe) · **confounder** (common cause of treatment and outcome) · **randomization** (kills confounding by design) · **identification** (the assumptions that make an effect recoverable) · **do-operator** (forcing a variable vs observing it) · **backdoor / collider** (what to control for, and what never to) · **ATE vs CATE** (one effect for all vs a different effect per customer).